

TidyTuesday Week 30

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Tuesday 21st July, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling installment of Two Random Time Series Walk Into a Regression, the segment where we let a computer play matchmaker between economic data that has never met and probably never should. This week the algorithm, in its infinite wisdom, decided to pair the Federal Reserve's holdings of medium-term Treasury securities with – and I cannot stress how excited you should be about this – the Consumer Price Index for Services in the South, Size Class D, a series so beloved that it has a popularity rating of 1 and was discontinued back in 2017. Truly, the Fed's bond portfolio has found its soulmate in a defunct regional price index. Somebody get these two a couples' registry.

And the raw regression came out swinging! An R-squared of 0.77, a p-value with more zeros after the decimal than I have the patience to count, the whole nine yards. If you stopped reading right here, you might march off convinced that the amount of Treasuries the Fed holds is somehow psychically linked to what Southerners were paying for services during the Carter-through-Obama era. It's a beautiful, statistically significant love story. It is also, and I say this with great affection, complete nonsense – because both of these series spend their entire lives politely drifting upward over time, and if you line up any two things that both go up, congratulations, you've discovered a "relationship." So has everyone who has ever noticed that both their age and the sea level increase.

This is exactly why we bother with the detrended version, where we rudely strip the shared upward march out of both series and see what's left. And what's left is... a considerable come-down. The R-squared face-plants from a swaggering 0.77 down to a much more sheepish 0.37, and the coefficient even flips its sign from positive to negative, which is data's way of saying "actually, never mind, forget I said anything." To be fair to our star-crossed pair, the detrended relationship is technically still significant, so this isn't a total mirage evaporating in the desert heat – but the drama has clearly been downgraded from "epic romance" to "two strangers who once made brief, confusing eye contact."

So what have we learned? Roughly nothing, and that's the whole point. These are randomly paired series with no causal story, no theoretical grounding, and absolutely no business being in the same spreadsheet – the strong raw number was mostly two trends holding hands, and even the surviving detrended blip is exactly the kind of coincidence you'd expect to trip over when you keep flinging random data at each other week after week. Spurious correlation isn't the bug here, it's the entire premise. Tune in next week, when the machine will pair something equally unrelated and I will once again pretend to be shocked.

2 Date: 2026-07-21

Series ID: TREAS1T5

This series is titled Assets: Securities Held Outright: U.S. Treasury Securities: Maturing in over 1 Year to 5 Years: Wednesday Level and has a frequency of Weekly, As of Wednesday. The units are Millions of U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2002-12-18 and the observation end date is 2026-07-15. The popularity of this series is 21.

Series ID: CUURD300SAS

This series is titled Consumer Price Index for All Urban Consumers: Services in South - Size Class D (DISCONTINUED) and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1977-12-01 and the observation end date is 2017-12-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CUURD300SAS	R-squared:	0.774			
Model:	OLS	Adj. R-squared:	0.765			
Method:	Least Squares	F-statistic:	82.38			
Date:	Tue, 21 Jul 2026	Prob (F-statistic):	3.15e-09			
Time:	13:11:10	Log-Likelihood:	-104.86			
No. Observations:	26	AIC:	213.7			
Df Residuals:	24	BIC:	216.2			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	214.5017	4.768	44.985	0.000	204.660	224.343
value_fred_TREAS1T5	6.405e-05	7.06e-06	9.076	0.000	4.95e-05	7.86e-05
	Omnibus:	1.003	Durbin-Watson:	0.294		
	Prob(Omnibus):	0.606	Jarque-Bera (JB):	0.795		
	Skew:	-0.004	Prob(JB):	0.672		
	Kurtosis:	2.143	Cond. No.	1.16e+06		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

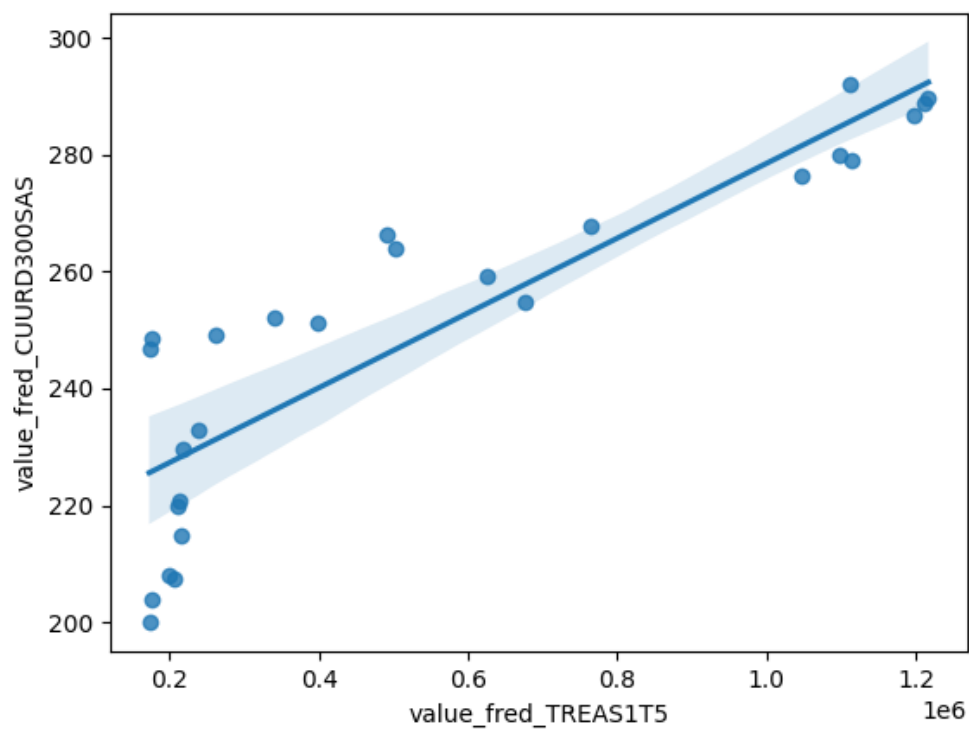


Figure 1: Raw Regression Plot for 2026-07-21

[2] The condition number is large, $1.16e+06$. This might indicate that there are strong multicollinearity or other numerical problems.

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CUURD300SAS_detrended	R-squared:	0.365
Model:	OLS	Adj. R-squared:	0.339
Method:	Least Squares	F-statistic:	13.82
Date:	Tue, 21 Jul 2026	Prob (F-statistic):	0.00107
Time:	13:11:11	Log-Likelihood:	-63.094
No. Observations:	26	AIC:	130.2
Df Residuals:	24	BIC:	132.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.318e-12	0.559	7.72e-12	1.000	-1.154	1.154
value_fred_TREAS1T5_detrended	-1.309e-05	3.52e-06	-3.718	0.001	-2.04e-05	-5.83e-06

Omnibus:	0.194	Durbin-Watson:	1.006
Prob(Omnibus):	0.907	Jarque-Bera (JB):	0.403
Skew:	0.028	Prob(JB):	0.818
Kurtosis:	2.393	Cond. No.	1.59e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.59e+05. This might indicate that there are strong multicollinearity or other numerical problems.

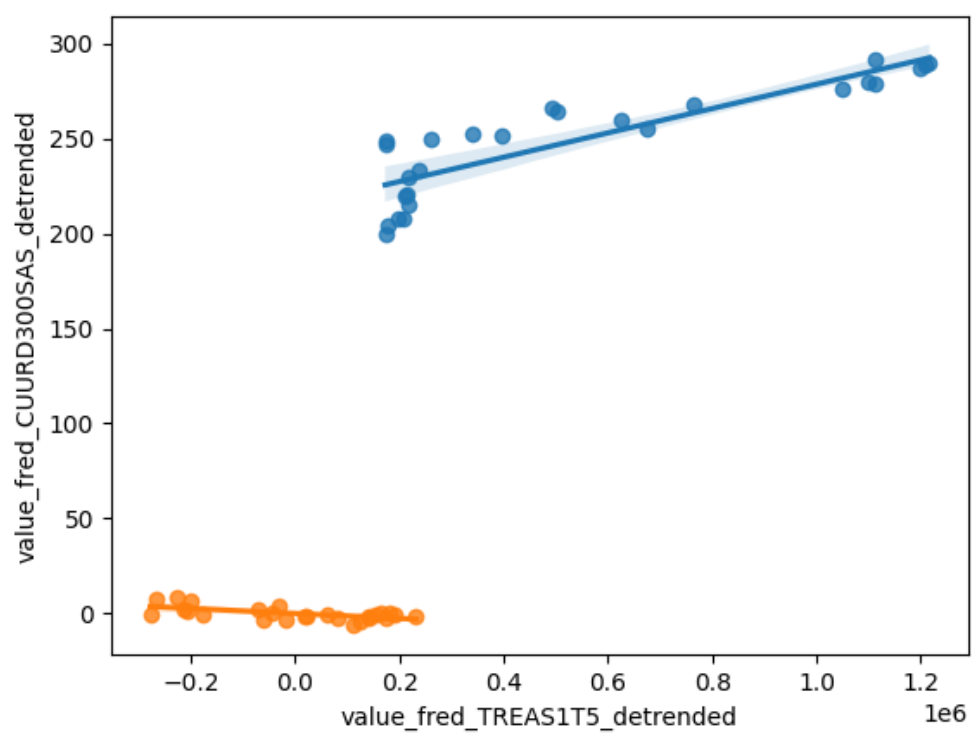


Figure 2: Detrended Regression Plot for 2026-07-21